

LAB 1: MICROSOFT WINDOWS INFRASTRUCTURE LAB WRITE-UP

CNIT242-LabSection3

Group 3

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BUSINESS SCENARIO

Applications used were Windows Server 2022 Disc Image, Windows 11, and Ventoy.

Applications used within Windows Server 2022 are Powershell, Server Manager, Active Directory (AD), “gpedit.msc” , Internet, and “secpol.msc”.

IP addresses assigned were 44.38.3.0/24 (last number ranging from 0 to 255). The 3 IP addresses used were 44.38.3.2 (member 1), 44.38.3.3 (member 2), and 44.38.3.4 (member 3).

The default IP address of DNS is 44.2.1.44 and the alternate one is 44.2.1.45. The IP addresses of the Time Servers assigned were “tick.cit.lcl”/“tock.cit.lcl”.

PROCEDURES

The procedures section incorporated step-by-step directions on how to replicate the tasks performed to complete the Microsoft Windows Administration Setup lab. Each section contains a numbered list showing the steps. Sections' headers, which are bolded and centered, showing the actions being taken on which workstations. by pre-install of the bootable Universal Serial Bus (USB), during install of bootable USB, Active Directory Domain Services Configuration, New Forest Creation. The followings are the formatting keys: **buttons** are bold, *options* are italicized, text entered into the computer is in Courier New, menu navigation is by the pipe symbol and italic words: *Start | Programs | MS Office | Word*.

Pre-install of bootable USB (All 3 members)

The following steps documented the steps of downloading software from the CIT network and transferring the files to a USB drive.

1. Inserted USB into laptop/personal computer (PC)
2. Navigated CIT-NET Lab Information Brightspace page | *Content | VPN Information & Files | Windows OpenVPN Client*
3. Clicked *Windows Installer with Configuration Files* to download OpenVPN Client and Graphical User Interface (GUI)
4. Navigated CIT-NET Lab Information Brightspace page | *Content | VPN Information & Files | OpenVPN Connect Configuration Files*
5. Clicked *CIT-Labs-OVPN-config* to download OpenVPN Zip File
6. Moved OpenVPN Zip File to the same folder as OpenVPN Client
7. Navigated *Show Hidden Icons* (On windows taskbar) and right-clicked *OpenVPN GUI*
8. Hovered over *CIT-Labs* and hit *Connect*

9. Typed Purdue career account username and password, then hit *Ok*
10. Clicked *File Explorer* | *Network* and typed \\rtfm.cit.lcl\Pub in search bar atop the page
11. Clicked *CNIT24200* folder
12. Checked box to select file *Network Drivers* | *ventoy-1.0.99* |
en-us_windows_11_consumer_editions_version_24h2_x64_dvd_1d5fcad3 |
en-us_windows_server_2022_updated_july_2023_x64_dvd_541692c3
13. Right-clicked selected files and clicked *Copy*
14. Navigated to *Lexar* (USB folder name) on left-hand side of file explorer
15. Pasted copied files to USB folder

During installation of bootable USB (on Dell OptiPlex 5060 workstation)

The following steps described booting the Dell OptiPlex 5060 workstation from the USB device using the UEFI boot menu to install Windows Server 2022 through Ventoy.

1. Plugged the USB into the Dell OptiPlex 5060 workstation
2. Powered the Dell OptiPlex 5060 workstation.
3. While the Dell OptiPlex 5060 workstation was booting, clicked **F12** to display the boot menu
4. Hit *UEFI: (USB Name)* from the *UEFI BOOT:* section
5. On Ventoy menu, hit

en-us-windows_server_2022_updated_july_2023_x64_dvd_541692c3.iso | *Boot in normal mode*

During installation of bootable USB (on Dell Pro Tower Plus workstations)

The following steps described booting the Dell Pro Tower Plus workstations from the USB device using the UEFI boot menu to install Windows Server 2022 through Ventoy.

1. Plugged the USB into the Dell Pro Tower Plus workstation
2. Powered the Dell Pro Tower Plus workstation.
3. While the Dell Pro Tower Plus workstation was booting, clicked **F12** to display the One-Time Boot Settings
4. Clicked *Windows Boot Manager* | *BIOS SETUP* | *Boot Configuration* | *UEFI USB* | *LOAD DEFAULTS* | *BIOS Defaults* | *OK* | *OK* | *OK* | *OK* under the UEFI Boot Devices section
5. Hit *en-us-windows_server_2022_updated_july_2023_x64_dvd_541692c3.iso* | *Boot in normal mode* on Ventoy menu

Active Directory Domain Services Configuration (All 3 workstations)

The following steps demonstrated the installation of Active Directory Domain Services, configuration of network settings, renaming each server accordingly, and resetting the administrator password.

1. Pressed **Windows**, typed *Server Manager* and pressed **Enter** to open Server Manager application
2. Clicked *Manage* | *Add Roles and Features* | *Next* | *Next* | *Next*
3. Checked *Active Directory Domain Services*
4. Clicked *Next* | *Next* | *Install* | *Close*
5. Once the installation was done, clicked *Local Server* | *Ethernet*
6. Right-clicked *Ethernet 2*. Hit *Disable*
7. Right-clicked *Ethernet*. Hit *Properties* | *Internet Protocol Version 4 (TCP/IPv4)*

8. Typed `255.255.255.0` in the *Subnet mask* field, `44.38.3.1` in the *Default gateway* field, `44.2.1.45` in the *Preferred DNS server* field, and `44.2.1.44` in the *Alternate DNS Server* field. For the *IP address* field, typed the member's IP number (see the BUSINESS SCENARIO section)
9. Clicked *OK* | *OK*
10. Opened Server Manager. Clicked *Tools* | *Active Directory Users and Computers* | *Users*
11. Right-clicked *Administrator*. Clicked *Reset Password...*
12. Typed a secure password in the *New password:* field and *Confirm password:* field.
Clicked *OK*
13. Opened Server Manager. Clicked *Local Server* | *Computer name* | *Computer Name* | *Change...* | *OK*
14. Typed `G03SRV01`, `G03SRV02`, and `G03SRV03` on workstation number 1, workstation number 2, and workstation number 3, respectively

DNS Configuration of the parent domain (on member 2 and 3's workstation)

The following steps documented the configuration of DNS settings and the creation of a new Active Directory forest.

1. Pressed **Windows**, typed *Server Manager* and pressed **Enter** to open Server Manager application
2. Clicked *Local Server* | *Ethernet*
3. Right-clicked *Ethernet*. Hit *Properties*
4. Disabled *Internet Protocol Version 6 (TCP/IPv6)*
5. Double-clicked *Internet Protocol Version 4 (TCP/IPv4)*

6. Typed 44.38.3.2 in the *Preferred DNS server* field, and 44.2.1.44 in the *Alternate DNS Server* field.
7. Clicked *OK* | *OK*

New Forest Creation (on Member 1's workstation)

The following steps to create a new forest for the parent domain, in which the child domains will be created within.

1. Pressed **Windows**, typed *Server Manager* and pressed **Enter** to open *Server Manager* application
2. Clicked “*flag icon*” | *Promote this server to a domain controller* | *Add a new forest*
3. Typed group03.c24200.cit.lcl into the *Root domain name* field.
4. Clicked *Next*.
5. Typed password. Recorded the password for future use.
6. Clicked *Next* | *Next*
7. Typed Group03 into *The NetBIOS domain name:* field
8. Clicked *Next* | *Next* | *Next* | *Install* | *Close*
9. Restarted the computer.

Adding a new domain into an existing forest (on Member 2's workstation)

The following steps were taken to promote the second server to a domain controller and as a child domain within the forest created by the designated parent domain.

1. Pressed **Windows**, typed *Server Manager* and pressed **Enter** to open *Server Manager* application
2. Clicked “*flag icon*” | *Promote this server to a domain controller* | *Add a new domain to an existing forest*

3. Typed `group03.c24200.cit.lcl` into the *Parent domain name* field
4. Typed `c242-03-b` into the *New domain name* field
5. Clicked *Select...*
6. Typed `GROUP03\Administrator` into the *User* field
7. Typed the password created in section 3b into the *Password* field
8. Clicked *OK* | *group03.c24200.cit.lcl* | *OK*
9. Typed a new password into the *Password:* field and *Confirm password:* field
10. Clicked *Next*.
11. Typed `GROUP03` into the *NetBISO domain name:* field.
12. Clicked *Next* | *Next* | *Next* | *Install* | *Close*
13. Restarted the computer

Adding a new domain into an existing forest (on Member 3's workstation)

The following steps were taken to promote the second server to a domain controller and as a child domain within the forest created by the designated parent domain.

1. Pressed **Windows**, typed `Server Manager` and pressed **Enter** to open Server Manager application
2. Clicked “*flag icon*” | *Promote this server to a domain controller* | *Add a new domain to an existing forest*
3. Typed `group03.c24200.cit.lcl` into the *Parent domain name* field
4. Typed `c242-03-a` into the *New domain name* field
5. Clicked *Select...*
6. Typed `GROUP03\Administrator` into the *User* field
7. Typed the password created in section 3b into the *Password* field

8. Clicked *OK* | *group03.c24200.cit.lcl* | *OK*
9. Typed the new password into the *Password:* and *Confirm password:* fields
10. Clicked *Next*.
11. Typed `GROUP03` into the *NetBISO domain name:* field.
12. Clicked *Next* | *Next* | *Next* | *Install* | *Close*
13. Restarted the computer

Remote Desktop Configuration (All 3 Workstations)

The following steps documented the configuration of Remote Desktop access by enabling Remote Desktop, adding the administrators to the allowed user list, and configuring the correct local security policies for the domain.

1. Pressed **Windows**, typed `Server Manager` and pressed **Enter** to open Server Manager application
2. Clicked *Local Server* | *Remote Desktop* | *Allow Remote Connections To This Computer*
3. Checked box, *Allow connections only from computers running Remote Desktop with NLA*
4. Clicked *Select Users* | *Add* | *Locations* | *Entire Directory* | *group03.c24200.cit.lcl* | *Users* | *Advanced* | *Find Now* | *Administrator* | *OK*
5. Clicked *Add* | *Locations* | *Entire Directory* | *group03.c24200.cit.lcl* | *c242-03-a/b.group03.c24200.cit.lcl* | *Users* | *Advanced* | *Find Now* | *Administrator* | *OK*
6. Pressed **Windows + R**
7. Typed `secpol.msc`
8. Clicked *Local Policies* | *User Rights Assignment* | *Allow Log On Through Remote Desktop Services*

9. Clicked *Locations* | *Entire Directory* | *group03.c24200.cit.lcl* | *Users* | *Advanced* | *Find now* | *Administrator* | *OK*
10. Pressed **Windows** + **r**
11. Typed `secpol.msc`
12. Clicked *Local Policies* | *User Rights Assignment* | *Allow Log On Through Remote Desktop Services*
13. Clicked *Locations* | *Entire Directory* | *group03.c24200.cit.lcl* | *c242-03-a/b.group03.c24200.cit.lcl* | *Users* | *Advanced* | *Find now* | *Administrator* | *OK*

Application Updates Configuration (All 3 workstations)

The following steps were taken to install required updates and ensure updates were installed automatically every 12 hours.

1. Pressed **Windows**, typed `Server Manager` and pressed **Enter** to open Server Manager application
2. Clicked *Local Server* | *Windows Updates* | *Install Now* | *Restart now*
3. Pressed **Windows** + **r**
4. Typed `gpedit.msc`
5. Clicked *Computer Configuration* | *Administrative Templates* | *Windows Components* | *Windows Update* | *Automatic Updates Detection Frequency*
6. Clicked *Enable*
7. Typed 12 into the *interval* field

Time Zone Configuration (All 3 workstations)

The following steps documented the necessary steps to sync the server time zone with the Purdue Computer Information Technology (CIT) provided time zone.

1. Pressed **Windows**
2. Typed `Powershell`
3. Pressed **Enter**
4. Typed `w32tm /config /manualpeerlist:"tick.cit.lcl" /syncfromflags:manual /reliable:YES /update`
5. Typed `w32tm /resync`
6. Clicked *Server Manager | Local Server | Time Zone | Change Time Zone*
7. Selected *Indiana (East)*
8. Clicked *OK | Apply | OK*

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